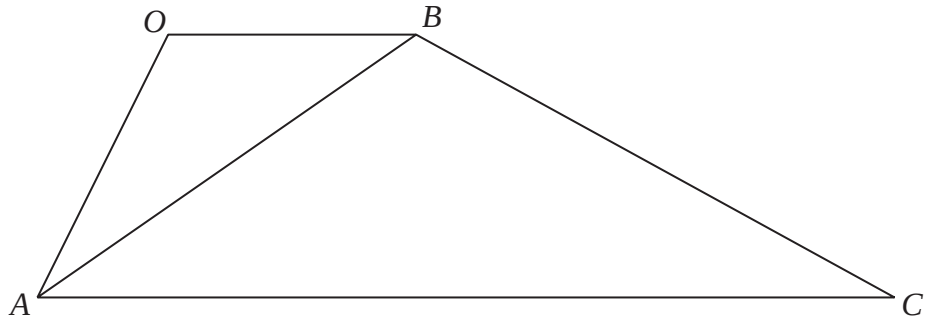
20



The diagram shows a trapezium $AOBC$.
AC is parallel to OB.
$AC = 4OB$.

$\overrightarrow{OA} = \mathbf{a}$ $\overrightarrow{OB} = \mathbf{b}$

The point P lies on the line AB such that $AP : PB = 4 : 1$

Using a vector method, prove that OPC is a straight line.

(Total for Question 20 is 5 marks)